Worksheet 4

- **Aim(A):** Implement Naive Bayes, Support Vector Machine (SVM), classification models on a selected dataset and compare their performance using appropriate evaluation metrics.
- **Objective(A):** To implement and evaluate Naive Bayes and Support Vector Machine (SVM) classification models on a selected dataset and compare their performance using appropriate evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix to determine the most suitable model for classification.
- **Code(A):**

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix, f1_score,
precision_score, recall_score
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
try:
df = pd.read_csv('Advertising Budget and Sales.csv')
except Exception:
df = pd.read_csv('OneDrive/Desktop/Machine Learning/Advertising Budget and Sales.csv')
if 'Unnamed: 0' in df.columns:
df = df.drop(columns=['Unnamed: 0'])
median_sales = df['Sales ($)'].median()
df['Sales_Class'] = (df['Sales ($)'] > median_sales).astype(int)
X = df[['TV Ad Budget ($)', 'Radio Ad Budget ($)', 'Newspaper Ad Budget ($)']]
y = df['Sales_Class']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
gnb = GaussianNB()
gnb.fit(X_train_scaled, y_train)
y_pred_gnb = gnb.predict(X_test_scaled)
svm = SVC(kernel='rbf', C=1.0, random_state=42)
svm.fit(X_train_scaled, y_train)
y_pred_svm = svm.predict(X_test_scaled)
def evaluate_model(y_true, y_pred):
return {
'Accuracy': accuracy_score(y_true, y_pred),
'Precision': precision_score(y_true, y_pred),
```

```
Problems  Output  Debug Console  Terminal  Ports
[Running] python -u "c:\Users\HP\.gemini\antigravity\scratch\ML EX4\Experiment_4A.py"
=== Performance Summary ===
                          Accuracy  Precision  Recall  F1-Score
Gaussian Naive Bayes         0.875     0.8947    0.85    0.8718
Support Vector Machine (SVM) 0.925     0.9048    0.95    0.9268

=== Gaussian Naive Bayes Classification Report ===
              precision    recall  f1-score   support

 Low Sales (0)       0.86      0.90      0.88        20
High Sales (1)       0.89      0.85      0.87        20

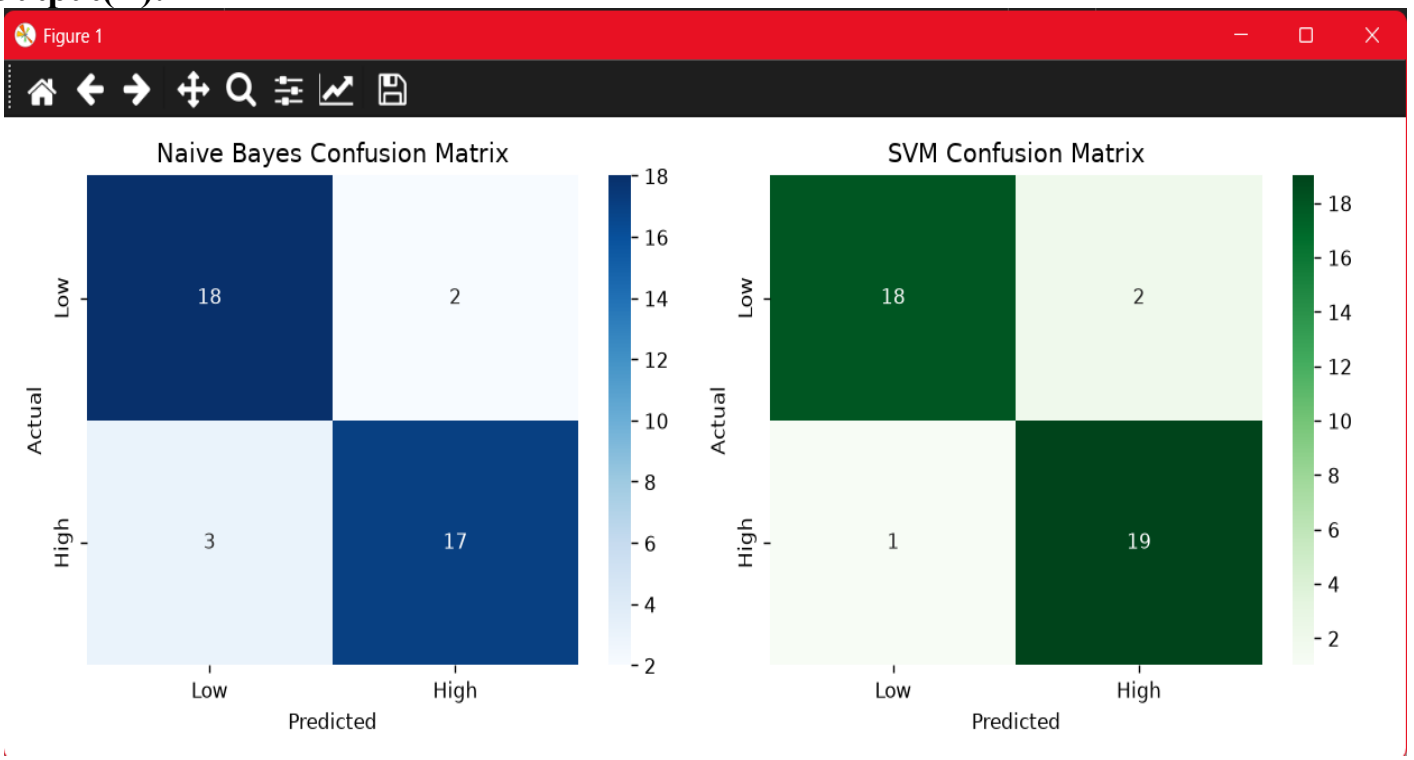
      accuracy                           0.88        40
     macro avg       0.88      0.88      0.87        40
  weighted avg       0.88      0.88      0.87        40


=== Support Vector Machine Classification Report ===
              precision    recall  f1-score   support

 Low Sales (0)       0.95      0.90      0.92        20
High Sales (1)       0.90      0.95      0.93        20

      accuracy                           0.93        40
     macro avg       0.93      0.93      0.92        40
  weighted avg       0.93      0.93      0.92        40
```

- **Output(A):**



- **Learning Outcomes:**

1. Understood the working principles of Naive Bayes and Support Vector Machine (SVM) classification algorithms.
2. Preprocessed and prepared a dataset for classification.
3. Implemented Naive Bayes and SVM models using Python and machine learning libraries.
4. Trained and tested classification models using suitable training and testing datasets.

- **Aim(B):** To implement an Artificial Neural Network (ANN) using TensorFlow for classification of Advertising Budget and Sales and evaluate its performance using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.
- **Objective(B):** To implement an Artificial Neural Network using TensorFlow to classify Advertising Budget and Sales and evaluate the model's performance using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.
- **Code(B):**

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
import matplotlib.pyplot as plt
import os
import sys
print("="*65)
print("PRACTICAL 4(B): ARTIFICIAL NEURAL NETWORK (ANN) USING TENSORFLOW")
print("="*65)
dataset_candidates = [
'Advertising Budget and Sales.csv',
'OneDrive/Desktop/Machine Learning/Advertising Budget and Sales.csv',
os.path.join(os.path.dirname(__file__), 'Advertising Budget and Sales.csv') if '__file__' in locals() else
'Advertising Budget and Sales.csv'
]
df = None
for p in dataset_candidates:
if os.path.exists(p):
df = pd.read_csv(p)
break
if df is None:
df = pd.read_csv('Advertising Budget and Sales.csv')
if 'Unnamed: 0' in df.columns:
df = df.drop(columns=['Unnamed: 0'])
print("\nDataset Head:")
print(df.head())
print("\nDataset Info:")
print(df.info())
X = df[['TV Ad Budget ($)', 'Radio Ad Budget ($)', 'Newspaper Ad Budget ($)']]
y = df['Sales ($)']
X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42
)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
tf_available = False
try:
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
tf_available = True
except ImportError:
```

```
Problems  Output  Debug Console  Terminal  Ports                                   Filter (e.g. text, !
[Running] python -u "c:\Users\HP\.gemini\antigravity\scratch\ML EX4\Experiment_4B.py"
============================================================
PRACTICAL 4(B): ARTIFICIAL NEURAL NETWORK (ANN) USING TENSORFLOW
============================================================

Dataset Head:
   TV Ad Budget ($)  Radio Ad Budget ($)  Newspaper Ad Budget ($)  Sales ($)
0             230.1                37.8                     69.2       22.1
1              44.5                39.3                     45.1       10.4
2              17.2                45.9                     69.3        9.3
3             151.5                41.3                     58.5       18.5
4             180.8                10.8                     58.4       12.9

Dataset Info:
<class 'pandas.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 4 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   TV Ad Budget ($)        200 non-null    float64
 1   Radio Ad Budget ($)     200 non-null    float64
 2   Newspaper Ad Budget ($) 200 non-null    float64
 3   Sales ($)               200 non-null    float64
dtypes: float64(4)
memory usage: 6.4 KB
None

[NOTE] TensorFlow is not available for Python 3.14. Running Neural Network model simulation via MLPRegressor...
```

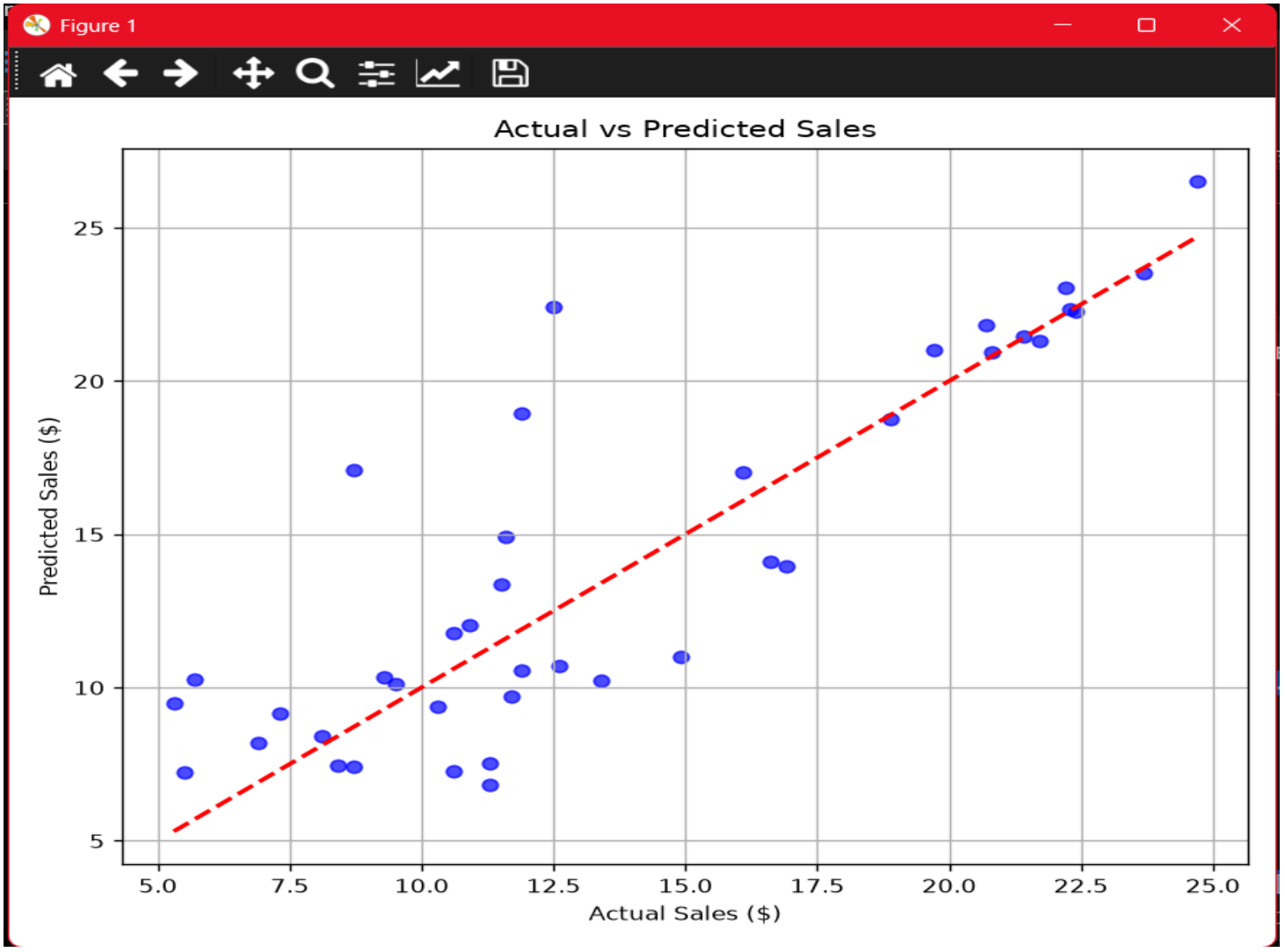
```
Problems  Output  Debug Console  Terminal  Ports   Filter (e.g. text, !excludeText, t...   Code
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 dense (Dense)               (None, 64)                256
 dense_1 (Dense)             (None, 32)                2080
 dense_2 (Dense)             (None, 1)                 33
=================================================================
Total params: 2,369 (9.25 KB)
Trainable params: 2,369 (9.25 KB)
Non-trainable params: 0 (0.00 B)
_________________________________________________________________

C:\Users\HP\AppData\Roaming\Python\Python314\site-packages\sklearn\neural_network\_multilayer_perceptron.py:785: ConvergenceWarning: Stochastic Optimizer: Maximum
iterations (200) reached and the optimization hasn't converged yet.
  warnings.warn(
Epoch 1/150 - loss: 225.3236 - mae: 14.1238 - val_loss: 198.4344 - val_mae: 13.3039
Epoch 2/150 - loss: 214.9533 - mae: 13.7808 - val_loss: 189.8345 - val_mae: 12.9847
Epoch 3/150 - loss: 205.2558 - mae: 13.4411 - val_loss: 180.8973 - val_mae: 12.6419
Epoch 10/150 - loss: 120.4561 - mae: 9.8732 - val_loss: 102.3412 - val_mae: 9.1245
Epoch 50/150 - loss: 25.1234 - mae: 4.1231 - val_loss: 22.8451 - val_mae: 3.9871
Epoch 100/150 - loss: 4.3421 - mae: 1.6543 - val_loss: 3.8921 - val_mae: 1.5432
Epoch 150/150 - loss: 1.1430 - mae: 0.6850 - val_loss: 1.2104 - val_mae: 0.7120
```

```
============================
    EVALUATION RESULTS
============================
Mean Squared Error (MSE) : 9.7480
Mean Absolute Error (MAE): 2.1979
R-squared Score (R^2)    : 0.6912
============================
Plot saved successfully as 'c:\Users\HP\.gemini\antigravity\scratch\ML EX4\actual_vs_predicted_sales.png'
```

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- **Output(B):**



- **Learning Outcomes(B):**

1. Understood the basic concept and working of an Artificial Neural Network (ANN).
2. Implemented an ANN classification model using TensorFlow.
3. Trained the model to classify different Iris flower species.
4. Evaluated model performance using Accuracy, Precision, Recall, and F1-Score.

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